Project Title

E-Commerce Customer Churn Prediction

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING (25SC2107E)

in

A.Y. 2026-2027

Submitted by

SAMEER REDDY(2520030276)

S. RIKHIL(2520030480)

Team Number: 35

Section: 9

Under the guidance of

Ms Rani Medidha

<Department of CSE>



Hyderabad-500090, Telangana, India.

ABSTRACT

Customer retention is essential for the long-term growth and revenue of modern e-commerce platforms, as acquiring new customers is significantly more expensive than retaining existing ones. Customer churn occurs when a user stops buying or browsing on the platform. This project presents a machine learning solution to predict which customers are likely to leave, enabling businesses to take proactive retention steps before churn happens.

The model analyzes key customer attributes, including account tenure, login behavior, order frequency, warehouse-to-home distance, satisfaction ratings, and recent complaints. The project implements a complete data pipeline involving missing value imputation, categorical encoding, and class imbalance handling. Multiple classification algorithms—such as Logistic Regression, Random Forest, and Gradient-Boosted Trees—are trained and compared to identify the most accurate predictor.

Model performance is evaluated using standard classification metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. In addition, feature importance analysis is applied to uncover the primary drivers behind customer churn. The resulting predictive pipeline offers actionable business intelligence, helping e-commerce companies reduce churn rates and design targeted customer retention strategies.

Signature of the Guide